Know the layers. Engineer the system.

A two-page guide to reading, practicing and proving AI systems engineering skills.

THE PURPOSE

Move beyond prompt use: explain **what executes at every layer**, build bounded systems, measure what changes, and defend choices with evidence. This manual is a learning route, not an instant-expert certificate.

46	L0-L9	16	64
PAGES	STACK LAYERS	STUDY WEEKS	CURATED RESOURCES

THE WHOLE STACK IN FOUR BOUNDARIES

L9	**INTERACTION** TUI, CLI, IDE, browser	Who sees and starts the task?
L8-L7	**CONTROL + EVIDENCE** Harness, agent loop, tools, MCP, RAG	Who chooses, permits and records actions?
L6-L5	**MODEL CONTRACT** API, chat template, tokenizer	What was actually sent to the model?
L4-L0	**EXECUTION** Inference, weights, kernels, OS, GPU	Where do time, memory and errors occur?

Critical distinction: The model predicts tokens; the runtime executes computations; the harness owns the task loop; the interface displays it. Security and observability span every layer.

READ THE MANUAL IN FOUR PASSES

01	**MAP** pp. 4-14	Stack, terminology, confusing concepts and system boundaries.
02	**UNDERSTAND** pp. 15-19	Models, tokenization, prefill/decode and local inference.
03	**BUILD** pp. 20-32	Agents, harness patterns, MCP, RAG, evaluation and security.
04	**PROVE** pp. 33-46	Trade-offs, capstones, weekly route, references and final exam.

PRACTICE PROTOCOL / EVIDENCE BEFORE CONFIDENCE

Turn reading into proof.

A repeatable process for every chapter - use it for terminology, diagrams, labs and design choices.

ONE STUDY SESSION / 65-90 MINUTES

01	**READ** 15 min	Identify the purpose, key terms, examples and open questions.
02	**TRACE** 10 min	Draw the request path. Label layer, owner, inputs and outputs.
03	**BUILD** 25-35 min	Run the smallest safe example. Capture configuration and a trace.
04	**COMPARE** 10-15 min	Change one variable; measure quality, time, memory or failure rate.
05	**EXPLAIN** 5-15 min	Teach it back; record trade-offs, negative tests and rollback.

SIX QUESTIONS TO ASK ABOUT EVERY NEW TERM

1. What problem does it solve?
2. Which layer owns it?
3. What enters and leaves?
4. What does it depend on?
5. What alternatives exist?
6. How would failure be proven?

16-WEEK PROGRESSION / THE MONTHLY CHECKPOINTS

MONTH 1
Explain tokens, transformers, inference and stack ownership.

MONTH 2
Operate a local model; build safe tools and bounded agent loops.

MONTH 3
Integrate RAG/MCP; evaluate and attack-test the workflow.

MONTH 4
Benchmark, harden, recover and defend a capstone design.

NEXT STEP / HTML TRAINING AND ASSESSMENT APP

Question coverage: At least 200 four-option, single-answer questions; at least 20 per assessed section, with at least half advanced. Suggested blueprint: 10 competency domains x 25 questions = 250 items (13+ advanced per domain).

Assessment rule: Randomize at least 20 distinct questions per exam, with balanced domain coverage and at least 50% advanced questions. Pass at **80% or higher** (16/20 on a 20-question exam); explain errors and direct learners back to the relevant chapter.

Use practice mode first, then section checks, then a mixed final. No HTML app is included in this PDF.